

An introduction to data science

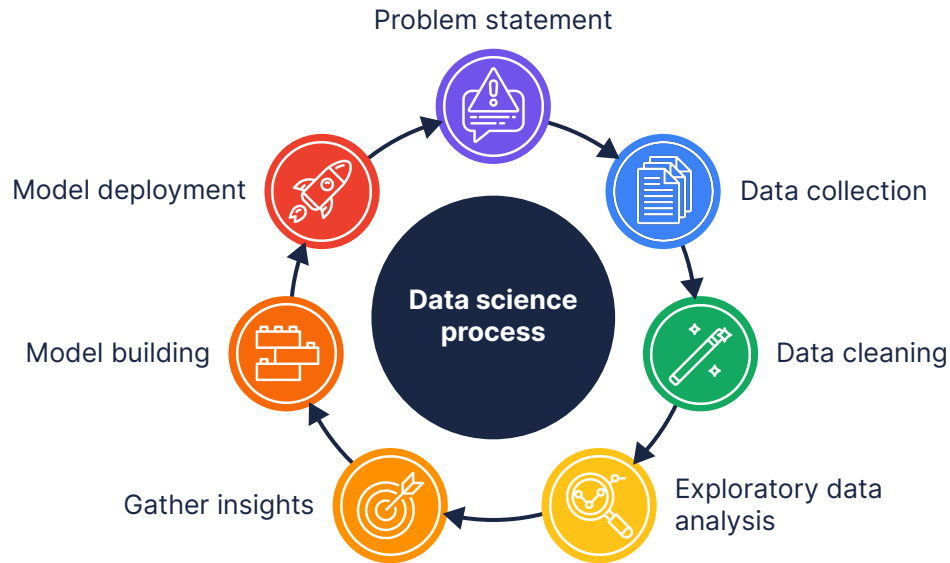
Approaches in data science

Overview

It is important that we are able to make **informed decisions** and **derive appropriate insights** from data.

We therefore need a **structured framework** for working with data and extracting valuable insights from it.

How our data science process is **applied** and **interpreted** depends on several factors, including whether we are doing a **quantitative** or **qualitative** analysis, and whether we need **hindsight**, **insight**, **foresight**, or **context**.



Quantitative and qualitative data analyses

Quantitative and qualitative data analyses are important because they enable us to gain a **more comprehensive understanding** of complex phenomena and make **data-driven decisions**.

Quantitative data analysis involves numerical measurement and statistical analysis.

It allows us to **measure and analyse numerical data** using **statistical methods**, enabling us to **identify patterns**, trends, and relationships between variables.

It is useful for making predictions, testing hypotheses, and identifying cause-and-effect relationships.

Qualitative data analysis involves exploring patterns and themes in non-numerical data.

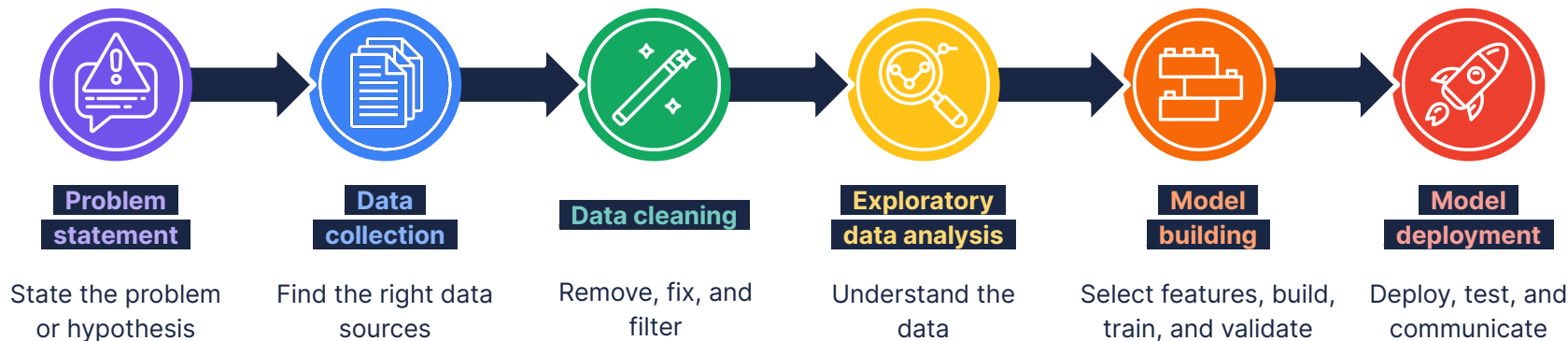
It allows us to **explore and interpret non-numerical data**, such as text, images, or videos.

It is useful for understanding the **context of a problem** and people's attitudes, behaviours, etc.

Both types of analysis are important because they provide **different ways of understanding** and **interpreting data**.

The data science process

The data science process is a systematic approach to **transforming a data problem** into a **data-driven solution**.



This approach to data science helps us to **discover meaningful** patterns, relationships, and trends and helps us develop **accurate** and **robust** models. **Various forms** of this process are used **across different data disciplines**, including data analytics, science, and engineering, under various names, such as OSEMN and CRISP-DM.

Problem statement



The problem statement helps us **define the scope and objectives** of our analysis and ensures that our insights are **relevant**.

A **problem statement** identifies the gap between the **current (problem) state** and the **desired (outcome) state**. It should be specific, brief, concise, clear, unbiased, and measurable.

A problem statement may also be in the form of a **hypothesis**, which is a **proposed cause and effect** for a particular phenomenon or problem which has not yet been proven correct.

Examples:

Statement: We need to report on estimated water and electricity income from different customer groups.

Hypothesis: The estimated water and electricity income from domestic customers are 30% lower than from other customers.

Question: How much water and electricity income can we expect from commercial customers per month?

Data collection



Data collection includes **identifying and acquiring applicable data sources**, both internally and externally, which can help answer the problem statement.

We can use company data or open-source data, or collect our own data **depending on the nature** of our **problem** and the **analysis** we would like to do.

Examples:

Data acquired from **surveys** such as market research and customer satisfaction surveys.

Queried data from **databases** or **APIs** (Application Programming Interfaces) such as sales data and employee information.

Downloaded data from **open sources** and **cloud repositories** such as general census data.

Data cleaning



Data cleaning, also known as **data wrangling**, involves **transforming raw data into usable formats**.

We can use **several cleaning techniques to ensure** that our data are indeed **accurate** and of the required **quality**.
If our data are inaccurate, so will our insights be.

Examples:

Using **spreadsheets** or a **programming language** to remove irrelevant observations, handle missing values, fix structural issues, etc.

Using **regular expressions** for pattern matching and replacing data.

Using **data visualisation tools** such as PowerBI or spreadsheets for identifying outliers and anomalies.

Exploratory data analysis



Exploratory data analysis (EDA) is an approach used to **summarise the main characteristics of a dataset** using aggregations, fundamental statistics, and visualisation techniques.

Before we can gather insights or build a model, we first need to **understand our data**. We can use **non-graphical methods**, such as descriptive statistics and correlation, or **graphical (visualisation) methods** to investigate our data.

Examples:

Descriptive statistics
Aggregations
Measures of central tendency
Measures of distribution
Correlation

Standard dev.
Count
Mean
Kurtosis
Pearson



Bar



Scatter



Density



Violin

Univariate and multivariate analyses

In EDA, we do either a univariate or multivariate analysis, depending on what we want to investigate.

Univariate analysis is the exploration of individual variables in a dataset, i.e. we only consider one variable at a time.

Non-graphical

We can use **descriptive statistics** such as the standard deviation, central tendency, and measures of distribution.

Graphical

We can use **visualisations** such as **histograms**, **density plots**, and **box plots** to understand the characteristics of a variable.

In a **multivariate analysis** we're more interested in the relationship between the different variables of our dataset.

Non-graphical

We use **correlation** to understand the strength and direction between variables.

Graphical

We can use **visualisations** such as **heatmaps**, **scatter plots**, and **pair plots** to investigate the relationship.

Gather insights



Gathering insights, also known as **data dissemination**, involves **gathering** and **reporting** the **insights** derived from the analysis.

Insights may be gathered in and reported to stakeholders through dashboards and reports that include text and data visualisations.

Examples:

Using **spreadsheets** or a **programming language** to summarise data and construct insights to form a report.

Using **data visualisation tools** such as PowerBI or spreadsheets to visualise and report the insights.

Model building



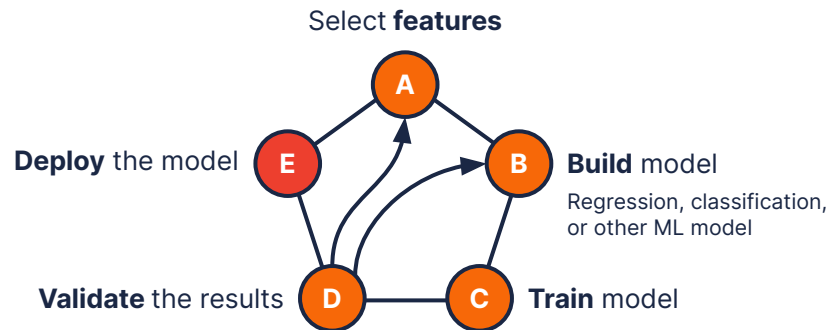
Model building involves **selecting an appropriate algorithm** and **training the model** on the data.

Model building often **involves reiteration** since a model will **rarely give us the results we seek on the first try**. This means that we train and test a model until we've found a suitable model before deploying it into a larger system.

Some common **tools** and **skills required for data collection** include:

Machine learning libraries
such as Scikit-learn and TensorFlow for building models in Python.

Deep learning libraries such as Keras and PyTorch for building neural networks in Python.



Model deployment



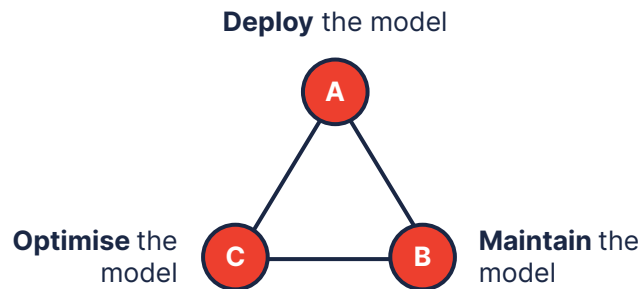
Model deployment involves **integrating the model** into a large system or application.

Deployment bridges the gap between data science and real-world applications. **Effective testing** and **communication** ensure the model is useful, reliable, and understood.

Although we have reached the end of the process, it is crucial to **maintain** and **optimise** the model.

Maintenance: Monitor and maintain the model, archiving insights to facilitate future endeavours.

Optimisation: Regularly retrain the model with new data sources and make adjustments to improve performance.



Type of analytics

The type of analytics we apply depends on our **goal** and prescribes our **approach** to the data analytics or data science process.

Descriptive

Hindsight

Used to describe **what** has happened in the **past**.

It's a summary of historical data that provides insights into patterns, trends, and relationships within the data

Examples: Dashboards and reports.

Diagnostic

Insight

Used to determine **why** something has happened in the **past**.

Helps organisations understand the factors that contributed to a particular outcome.

Examples: Data mining and drill-down analysis.

Predictive

Foresight

Used to forecast **what** will happen in the **future**.

Uses statistical models and machine learning algorithms to identify patterns and trends in historical data to predict future outcomes.

Examples: Forecasting and risk modelling.

Prescriptive

Context

Used to recommend the best **course of action** for a given situation.

Uses advanced algorithms and optimisation techniques to suggest the most optimal solution based on a variety of factors and constraints.

Examples: Optimisation